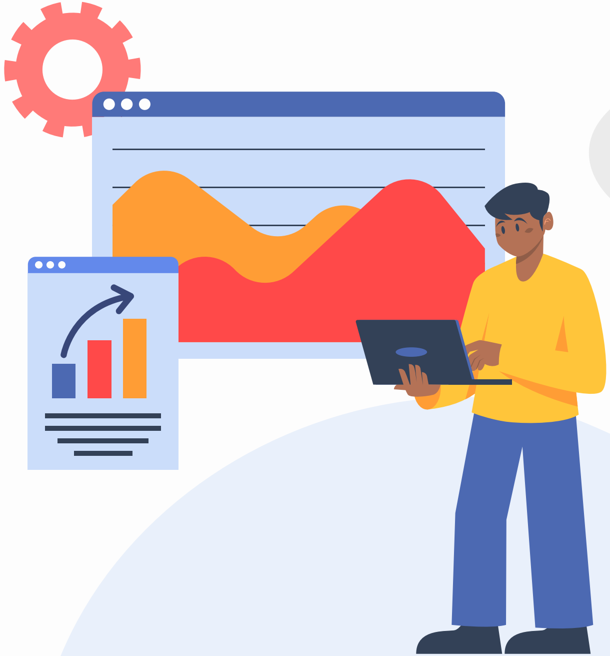
July 17th, 2026

BWSI Medlytics

Sleep Analysis Challenge Project Group 2



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Introduction

- Sleep disorders, such as impaired memory and learning, obesity, hypotension, and more can occur from inadequate sleep
    - 50-70 million adults in the US
- By analyzing data, experts can determine which sleep stage patients are in their sleep, which can help with diagnosing and treating the disorder
    - Our model with this prediction
- Collection of data can be costly, so high accuracy predicting models are necessary

Research Question

Can physiological signals recorded during sleep be used to classify what stage of sleep a person is in?

Preprocessing & Feature Extraction

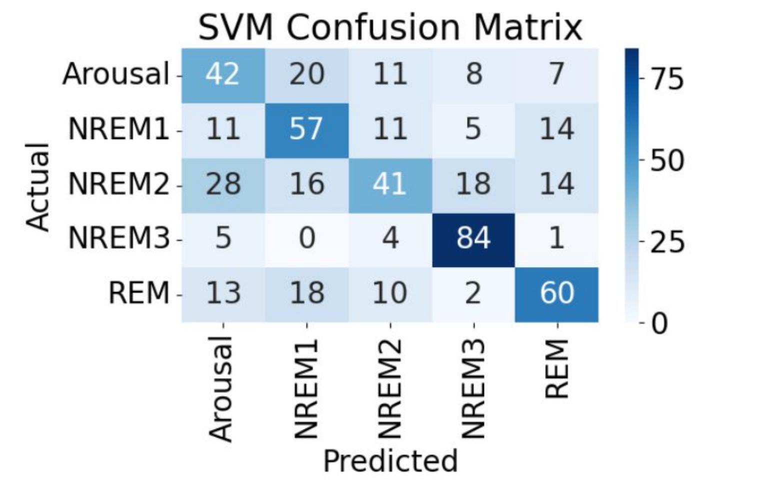
- Shuffle & Partition - 2500 training samples, 1000 validation, 500 testing
- FFT Transformation
  - Converted 60-second time-domain signals into the frequency domain using Fast Fourier Transform for 4 key signals EEG, EOG, Airflow, and ECG
- Spectral Band Summation

Delta	0.5-4 Hz
Theta	4-8 Hz
Alpha	8-13 Hz
Beta	13-30 Hz

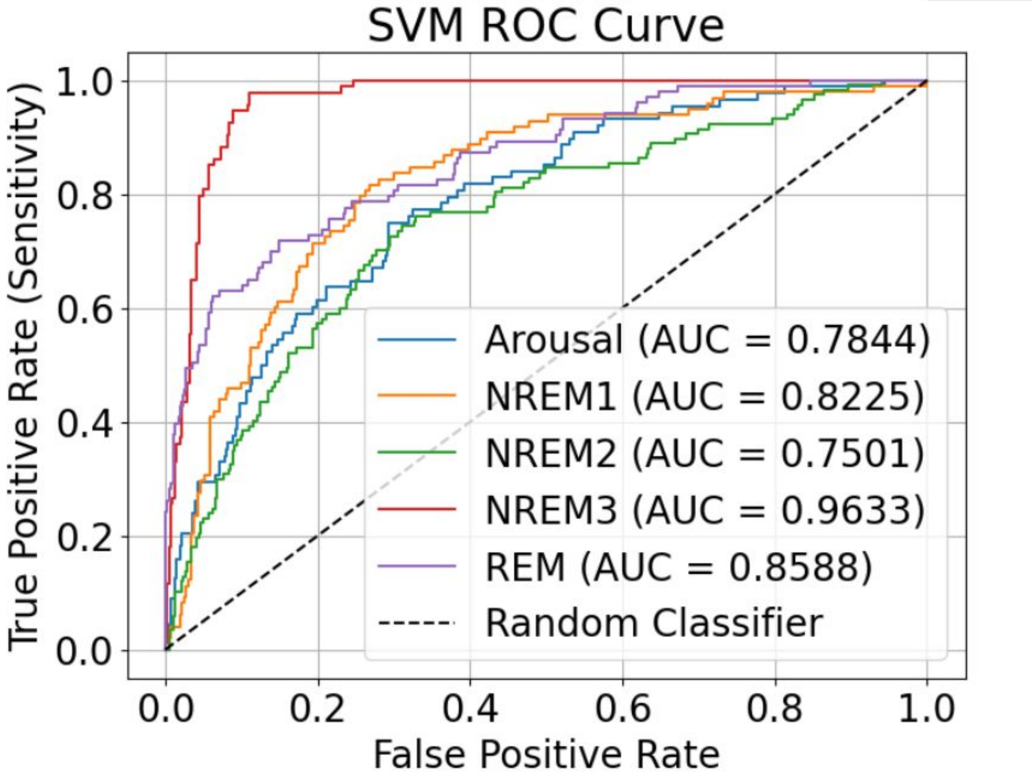
- Standardized features

Model #1 – SVM

Training Accuracy: 62.40%
Testing Accuracy: 56.80%

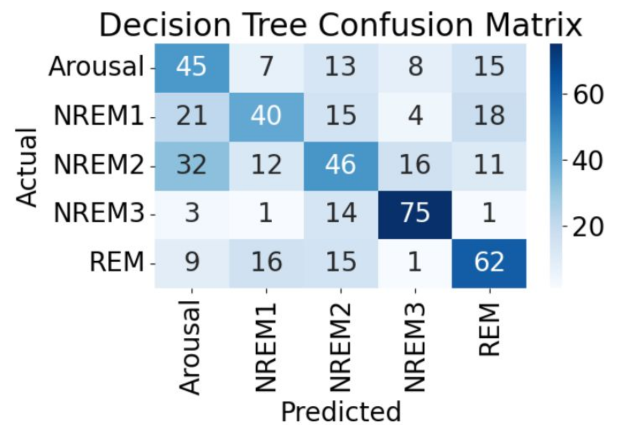


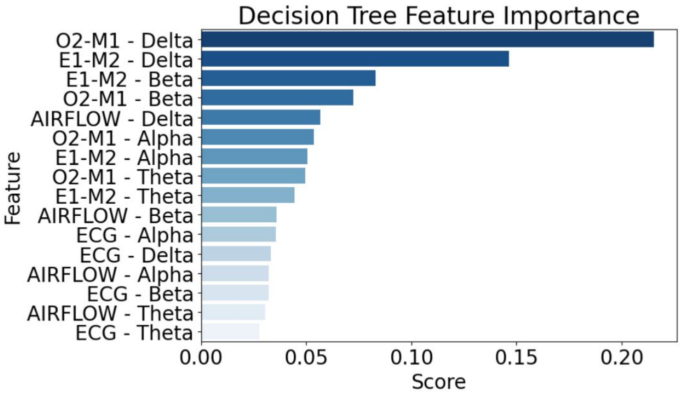
	precision	recall	f1-score	support
Arousal	0.42	0.48	0.45	88
NREM1	0.51	0.58	0.55	98
NREM2	0.53	0.35	0.42	117
NREM3	0.72	0.89	0.80	94
REM	0.62	0.58	0.60	103
accuracy			0.57	500
macro avg	0.56	0.58	0.56	500
weighted avg	0.56	0.57	0.56	500

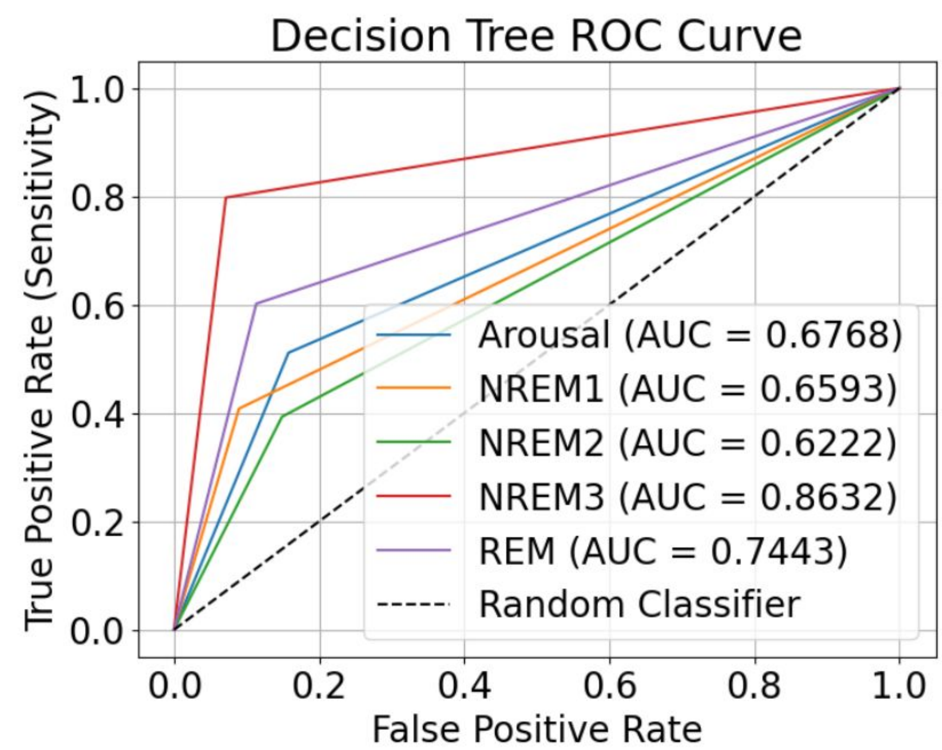


Model #2 – Decision Tree

Training Accuracy: 100%
Testing Accuracy: 53.60%

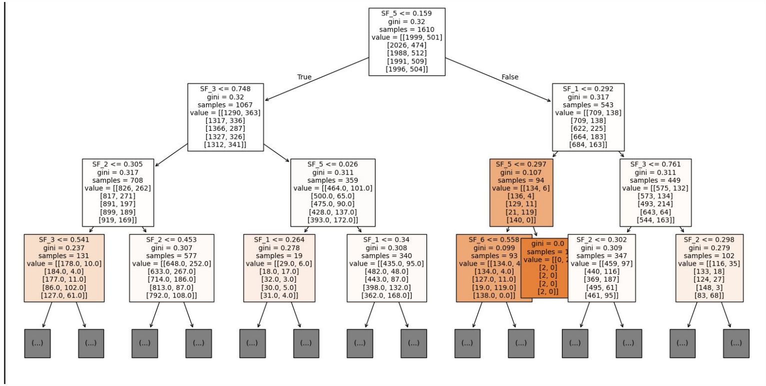


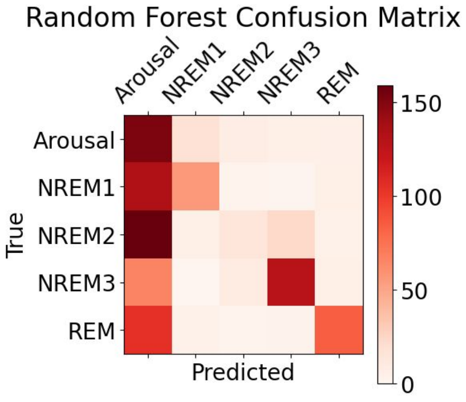


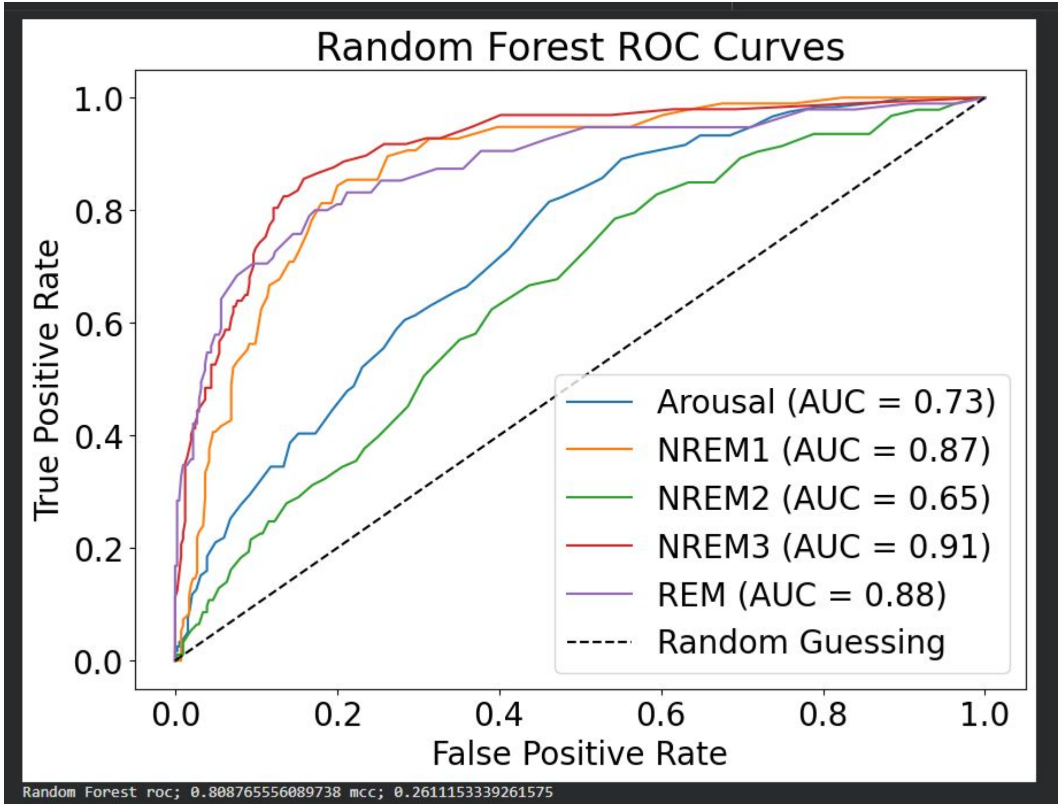


Model #3 - Random Forest

Training accuracy: 100.00%
Validation accuracy: 30.00%



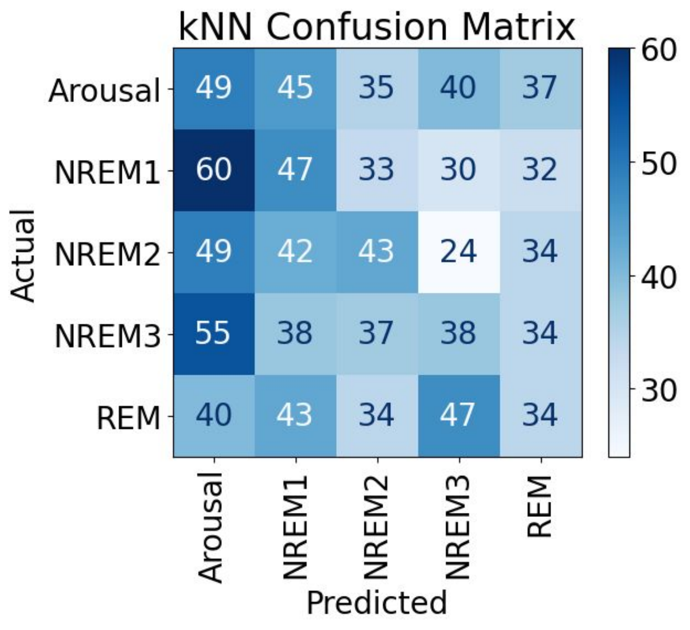




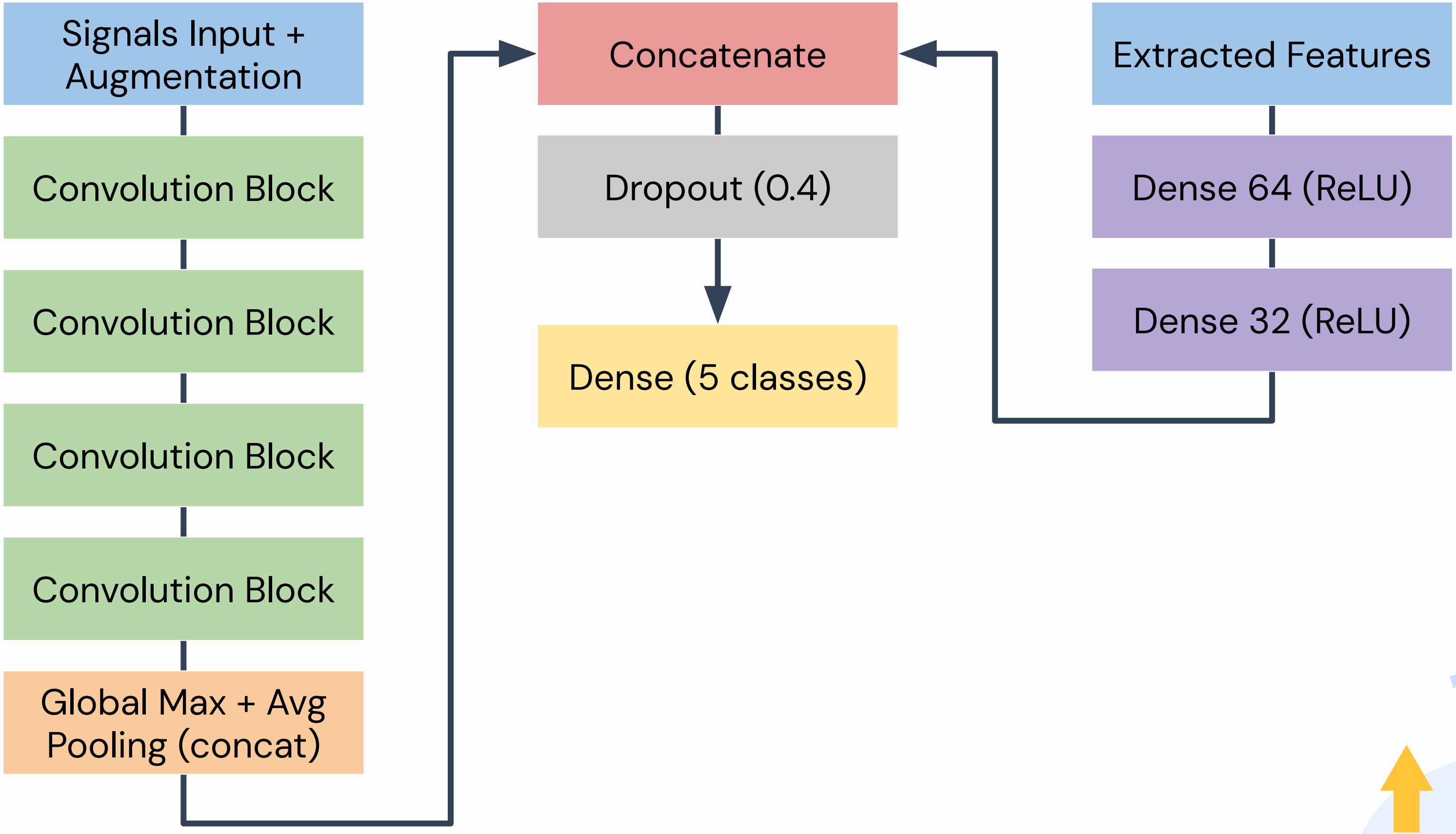
Test Model – kNN

Validation Accuracy: 0.211
Training Accuracy: 0.3492
Testing Accuracy: 0.214

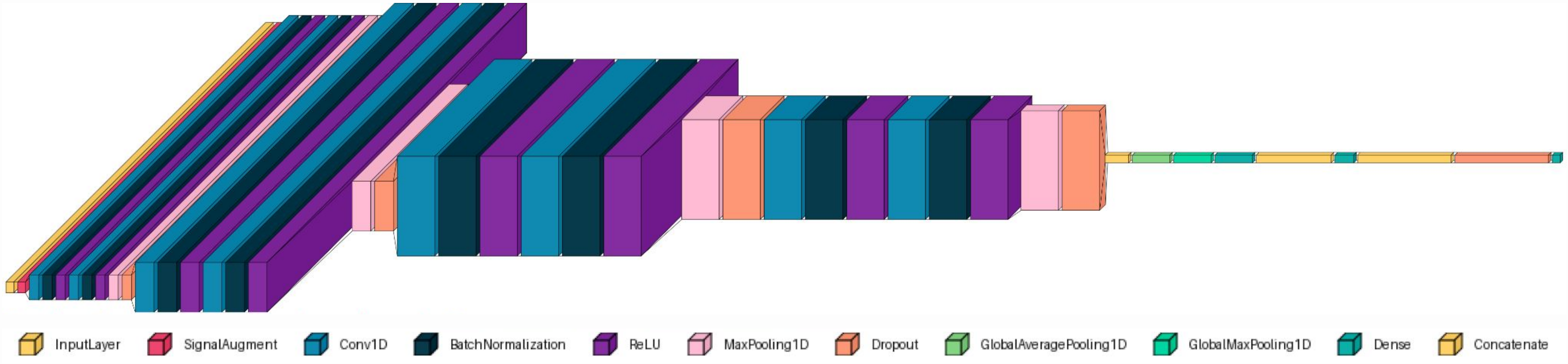
- Both training and testing accuracy were low
- The low validation accuracy also shows the inaccurate predictions of the model
- It is better to use a neural network model for this dataset to have better predictions for the sleep stage that an individual is in because neural networks are able to analyze more complex patterns



CNN – Structure

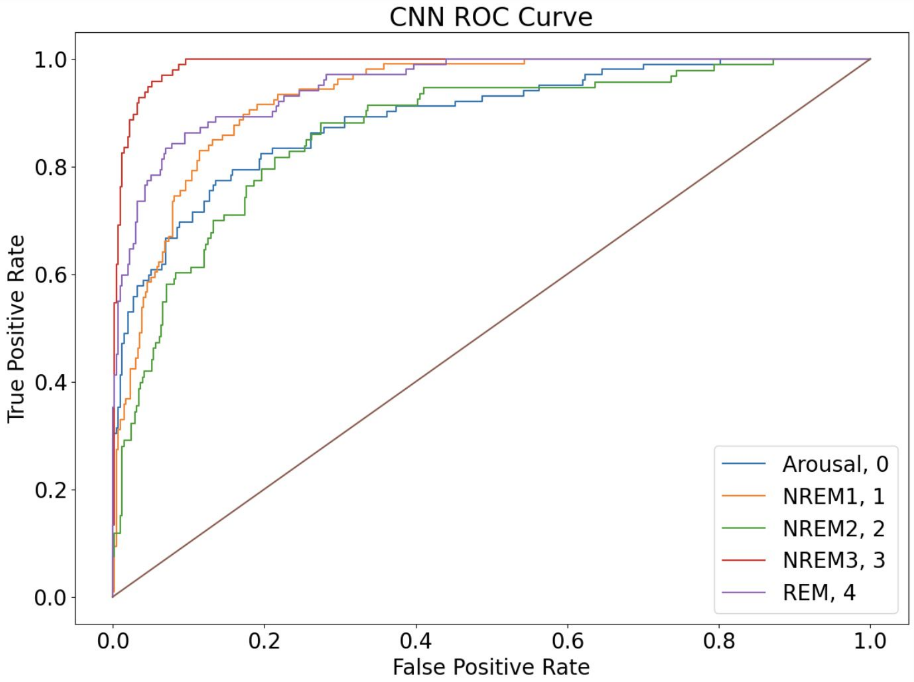


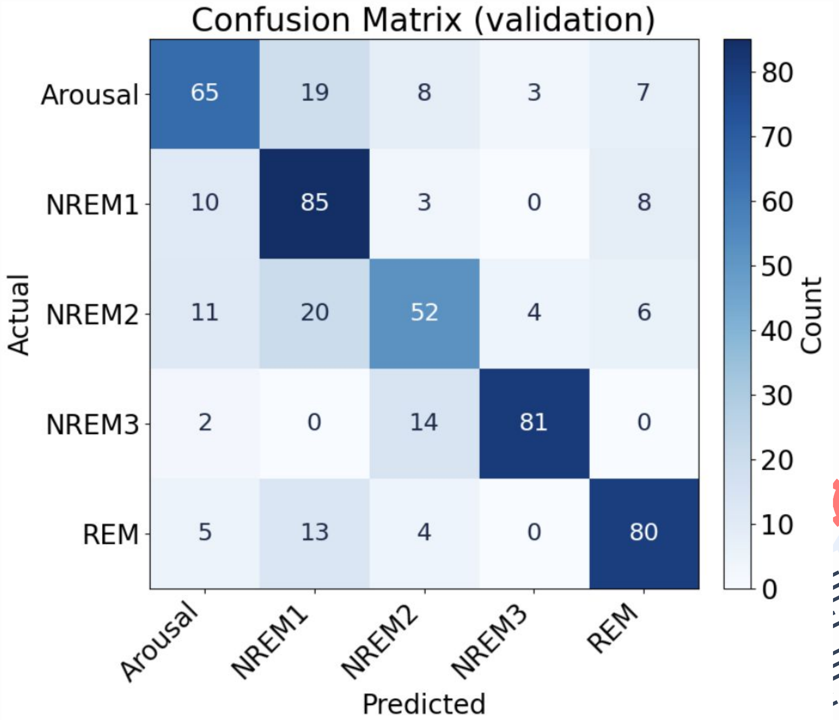
CNN – Structure



CNN – Results

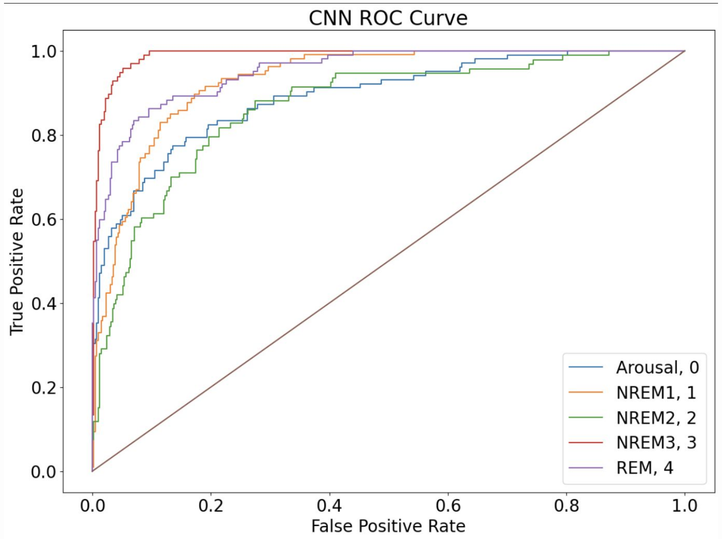
Training accuracy: 0.799
Validation accuracy: 0.726
ROC-AUC (val): 0.933
MCC (val): 0.587

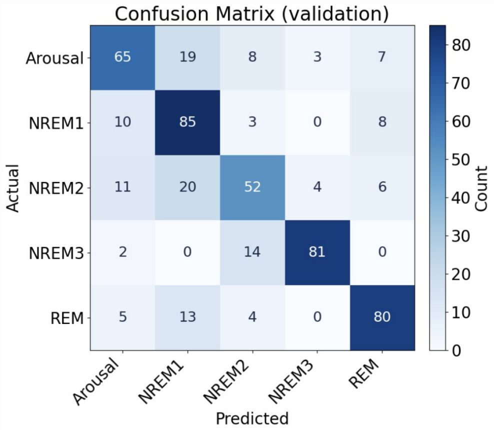




Results

- CNN had the highest accuracy and ROC-AUC on the validation set
  - Very accurate at identifying NREM1, NREM3, and REM sleep
  - NREM2 and arousal were the most difficult
  - Traditional models were far less accurate





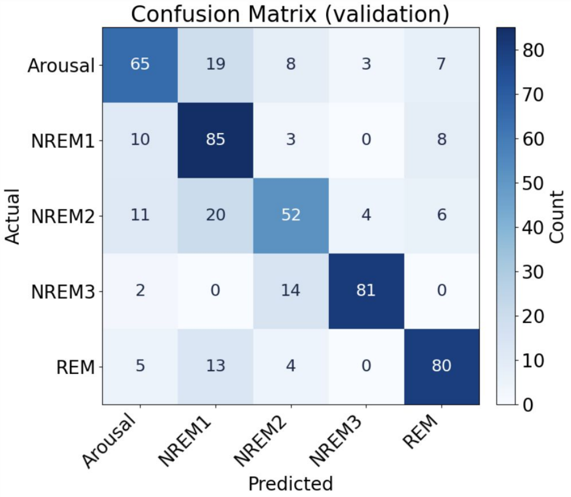
Interpretation/discussion

- Overall, the CNN can make strong predictions about sleep stages
  - Can be used to help monitor sleep health
  - Useful for diagnosing sleep disorders
- Complicated patterns are harder for traditional models to learn
  - Extracting useful features boosted model performance
  - Raw signal inputs can only help so much, and weren’t really useful to traditional models

Limitations

- Many of our models experienced forms of overfitting
- We had a small sample sizes with only around 4000 total training samples split across train/validation/test
- Other features to extract from signals to boost performance
- Training time for CNN made it slow to test new changes

Conclusion



Training accuracy: 0.799
Validation accuracy: 0.726
ROC-AUC (val): 0.933
MCC (val): 0.587

We can accurately predict which sleep stage a patient is in using a supervised machine learning model trained on physiological data, with the CNN model yielding decent accuracy and sensitivity, significantly performing other models like SVM, decision tree, and random forest, showing why a neural network is optimal, with high accuracy around NREM1 and NREM3.